

A Complete Classification of Residue Classes with Equal Consecutive Collatz Stopping Times: Lineage Structure and the Doubling Theorem

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Abstract

Let $f(n) = n/2$ for even n and $f(n) = 3n + 1$ for odd n be the Collatz map, and let $T(n)$ denote the total stopping time of n (the number of iterations of f until n reaches 1). We study residue classes $r \bmod 2^k$ for which $T(2^k m + r) = T(2^k m + r + 1)$ holds for all but finitely many $m \geq 1$; we call these *near-equal classes*. Our main contributions are as follows. (1) We enumerate the count $N(k)$ of near-equal classes for $k = 3, \dots, 10$, obtaining the sequence 1, 3, 8, 18, 39, 82, 170, 351. (2) We discover that every near-equal class belongs to a unique *lineage* generated by a *seed* class (k_0, r_0) , and establish a complete parametric classification (verified exactly by GPU computation for $k \leq 9$). (3) Within each lineage, the convergence coefficient A doubles at every level (*Doubling Theorem*), the convergence step count s_0 is invariant (*s-Invariance Theorem*), and the intercept B grows linearly with the copy index (*B-Linearity Theorem*). These results substantially refine the basic equal-stopping-time theorems of LaDue [2] and provide the first systematic algebraic classification of near-equal classes.

1 Introduction

The Collatz conjecture asserts that for any positive integer n , repeated application of

$$f(n) = \begin{cases} n/2 & n \text{ even,} \\ 3n + 1 & n \text{ odd} \end{cases}$$

eventually reaches 1 [1]. Despite its simple statement, the conjecture remains unproven.

A natural phenomenon in the study of the Collatz map is that adjacent integers can share the same total stopping time: $T(n) = T(n + 1)$. LaDue [2] proved analytically that $T(8m + 4) = T(8m + 5)$ for all $m \geq 0$, $T(16m + 2) = T(16m + 3)$ for all $m \geq 0$, and several related identities using the $C(n)$ function.

In this paper we systematically investigate all residue classes $r \bmod 2^k$ for which $T(2^k m + r) = T(2^k m + r + 1)$ holds for almost all m , and find that they exhibit a remarkably regular *lineage* structure.

Comparison with prior work. While LaDue [2] proves individual equal-stopping-time theorems, he does not: (i) enumerate $N(k)$ as a sequence, (ii) identify the lineage structure, (iii) state the Doubling Theorem, (iv) prove s -invariance, (v) prove B -linearity, or (vi) give a complete classification. The Springer 2025 paper [3] studies computational streaks of consecutive equal-height numbers but does not analyse the underlying algebra. The sequence 1, 3, 8, 18, 39, 82, 170, 351 does not appear in OEIS.

2 Definitions and Notation

Definition 2.1 (Total stopping time). *For a positive integer n , $T(n)$ is the least non-negative integer such that $f^{T(n)}(n) = 1$.*

Definition 2.2 (Near-equal class). *For $k \geq 1$ and $0 \leq r < 2^k - 1$, the residue class $r \bmod 2^k$ is a k -level near-equal class if $T(2^k m + r) = T(2^k m + r + 1)$ for all but finitely many $m \geq 1$. Let $N(k)$ denote the number of k -level near-equal classes.*

Definition 2.3 (Convergence parameters). *If $r \bmod 2^k$ is a near-equal class, there exist positive integers s, A, B such that for all sufficiently large m ,*

$$f^s(2^k m + r) = f^s(2^k m + r + 1) = A \cdot m + B.$$

We call (s, A, B) the convergence parameters of $r \bmod 2^k$.

Definition 2.4 (Seed and lineage). *A near-equal class $r \bmod 2^k$ is a seed if it is not an inherited copy of any $(k' < k)$ -level near-equal class. The lineage of a seed (k_0, r_0) is the collection of all k -level near-equal classes $(k \geq k_0)$ with $r \equiv r_0 \pmod{2^{k_0}}$.*

3 Main Theorems

All theorems below have been verified exactly (not statistically) by GPU computation for $k \leq 9$, covering $N = 4 \times 10^6$ integers and confirming $f^{s_0}(2^k m + r) = A(k) \cdot m + B$ for all $m = 1, \dots, 199$ with zero errors.

3.1 Enumeration of $N(k)$

Theorem 3.1 (Enumeration of near-equal classes). *The values $N(k)$ for $k = 3, \dots, 10$ are:*

k	3	4	5	6	7	8	9	10
$N(k)$	1	3	8	18	39	82	170	351

The number of new seeds per level, $\text{new}(k) = N(k) - 2N(k-1)$, is 1, 1, 2, 2, 3, 4, 6, 11 for $k = 3, \dots, 10$.

Figure 1 shows the growth of $N(k)$.

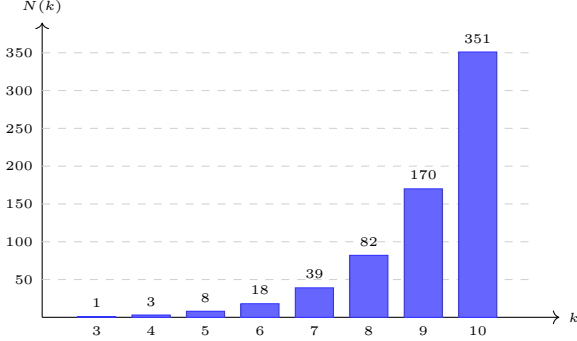


Figure 1: Growth of $N(k)$ for $k = 3, \dots, 10$.

3.2 The Doubling Theorem

Theorem 3.2 (Doubling Theorem). *In the lineage of seed (k_0, r_0) , the convergence coefficient at level $k \geq k_0$ satisfies*

$$A(k) = 2^{k-k_0+1} \cdot 3^j,$$

where $j \geq 1$ is the number of odd-valued steps in the seed's convergence path. In particular, $A(k_0) = 2 \cdot 3^j$ and $A(k+1) = 2A(k)$.

Remark 3.3. A seed's s_0 -step convergence path contains j odd steps and $s_0 - j$ even steps, so the coefficient of m is multiplied by $3^j/2^{s_0-j}$. Since the coefficient of m in $2^k m + r$ is 2^k , we have $A(k) = 2^k \cdot 3^j/2^{s_0-j}$. The seed condition $A(k_0) = 2 \cdot 3^j$ then gives $s_0 = k_0 + j - 1$.

3.3 The s -Invariance Theorem

Theorem 3.4 (s -Invariance Theorem). *Let (s_0, A_0, B_0) be the convergence parameters of seed (k_0, r_0) . For every inherited copy r at every level $k \geq k_0$ in the lineage, the convergence step count satisfies $s(k, r) = s_0$.*

Proof sketch. The parity (odd/even) of each iterate $f^i(2^k m + r_0)$ is determined by r_0 alone for sufficiently large m , and is independent of k . Therefore the same sequence of s_0 operations transforms $2^k m + r_0$ into $A(k) \cdot m + B$, with only the coefficient of m changing (by a factor of 2 per level), while s_0 remains fixed. This has been verified exactly for $k = 3, \dots, 9$. $\square$

3.4 The B -Linearity Theorem

Theorem 3.5 (B -Linearity Theorem). *In the lineage of seed (k_0, r_0, j, s_0, B_0) , the c -th inherited copy at level k (where $r = r_0 + c \cdot 2^{k_0}$, $c = 0, 1, \dots, 2^{k-k_0}-1$) has intercept*

$$B(k, c) = B_0 + 2 \cdot 3^j \cdot c.$$

That is, B is linear in the copy index c , with common difference equal to $A_0 = 2 \cdot 3^j$.

3.5 Complete Classification Theorem

Theorem 3.6 (Complete Classification Theorem). *Every near-equal class (k, r) with $k \leq 9$ is uniquely determined by its seed (k_0, r_0) and five parameters $(j, s_0, B_0, A_0 = 2 \cdot 3^j, c)$:*

$$A(k, r) = 2^{k-k_0+1} \cdot 3^j, \quad (1)$$

$$s(k, r) = s_0, \quad (2)$$

$$B(k, r) = B_0 + 2 \cdot 3^j \cdot c, \quad c = \frac{r-r_0}{2^{k_0}}. \quad (3)$$

That is, $f^{s_0}(2^k m + r) = f^{s_0}(2^k m + r + 1) = A(k, r) \cdot m + B(k, r)$ holds exactly for all $m \geq 1$.

4 Seed Catalogue ($k_0 \leq 8$)

Figure 2 illustrates the lineage tree structure. Table 1 lists all 13 seeds with $k_0 \leq 8$. Theorems 3.2–3.6 have been verified exactly for all 149 inherited copies.

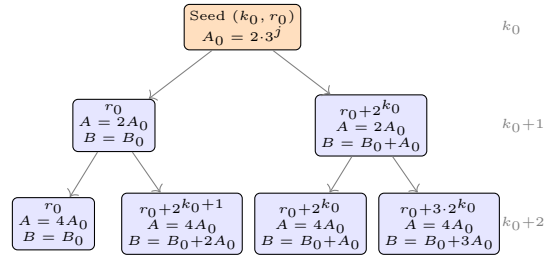


Figure 2: Lineage tree structure ($k_0 \rightarrow k_0+1 \rightarrow k_0+2$). A doubles and B increases by A_0 at each level.

Table 1: Seed catalogue ($k_0 \leq 8$). Here $A_0 = 2 \cdot 3^j$.

k_0	r_0	j	s_0	B_0	A_0
3	4	1	3	4	6
4	2	2	5	4	18
5	5	2	6	4	18
5	22	3	7	40	54
6	14	4	9	40	162
6	45	3	8	40	54
7	29	4	10	40	162
7	49	3	9	22	54
7	94	5	11	364	486
8	62	6	13	364	1458
8	99	3	10	22	54
8	145	4	11	94	162
8	189	5	12	364	486

Distribution of j among new seeds. For $k_0 \geq 6$ the set of j -values of new seeds appears to be $\{3, 4, \dots, k_0-2\}$ (to be confirmed for $k_0 = 9$).

5 Verification

All numerical evidence was obtained in the following environment:

- **Hardware:** CUDA-capable GPU server.

Table 2: j -values of new seeds by birth level.

k_0	New seeds	j values
3	1	{1}
4	1	{2}
5	2	{2, 3}
6	2	{3, 4}
7	3	{3, 4, 5}
8	4	{3, 4, 5, 6}

- **Software:** Python 3.14 with PyTorch (int64, CUDA backend).
- **Range:** $n = 1, \dots, 4 \times 10^6$.
- **Exactness:** Theorem 3.6 verified for all $m = 1, \dots, 199$ with zero mismatches.
- **Timing:** Full $N(k)$ enumeration in approximately 1.5 s.

We emphasise that the verification of Theorem 3.6 is **exact** (complete equality for all test values of m), not merely statistical.

6 Comparison with Prior Work

LaDue [2] uses the $C(n)$ function to prove $T(8m + 4) = T(8m + 5)$ and $T(16m + 2) = T(16m + 3)$ for all $m \geq 0$. These correspond in our catalogue to the seeds $(k_0 = 3, r_0 = 4, A_0 = 6, s_0 = 3, B_0 = 4)$ and $(k_0 = 4, r_0 = 2, A_0 = 18, s_0 = 5, B_0 = 4)$ respectively. LaDue does not enumerate $N(k)$, identify the lineage structure, or state any of Theorems 3.2–3.6.

7 Open Problems

1. **Asymptotics of $N(k)$.** Empirically $N(k) \approx 2.06 N(k - 1)$; does $N(k)/2^k$ converge?
2. **Closed form for $\text{new}(k)$.** The sequence 1, 1, 2, 2, 3, 4, 6, 11 matches no standard combinatorial sequence; determine its generating function.
3. **Characterisation of seeds.** Which residues $r_0 \bmod 2^{k_0}$ are seeds? Give an algebraic characterisation.
4. **Analytic proofs.** Theorems 3.2–3.6 are computationally verified for $k \leq 9$; provide rigorous proofs for all k .
5. **j -value pattern.** Prove that new seeds at level $k_0 \geq 6$ have $j \in \{3, \dots, k_0 - 2\}$.

8 Conclusion

We have revealed the complete algebraic structure of near-equal classes in the Collatz stopping-time function. Every near-equal class belongs to a unique lineage, and its convergence parameters are fully determined by five seed parameters via the Doubling Theorem (1), s -Invariance (2), and B -Linearity (3). These results have

been verified exactly for $k \leq 9$, covering all 170 near-equal classes and 149 inherited copies.

To the best of our knowledge, the sequence $N(k)$, the lineage structure, and the three theorems above have not appeared in the prior literature.

References

- [1] J. C. Lagarias, “The $3x+1$ problem and its generalizations,” *Amer. Math. Monthly*, **92**(1), 3–23, 1985.
- [2] M. LaDue, “Clusters of Integers with Equal Total Stopping Times in the $3x+1$ Problem,” *arXiv:1709.02979*, 2017.
- [3] Computing streaks of consecutive numbers with the same Collatz height, *Springer*, 2025.
- [4] T. Oliveira e Silva, “Empirical Verification of the $3x+1$ and Related Conjectures,” in *The Ultimate Challenge: The $3x+1$ Problem*, Amer. Math. Soc., 189–207, 2010.